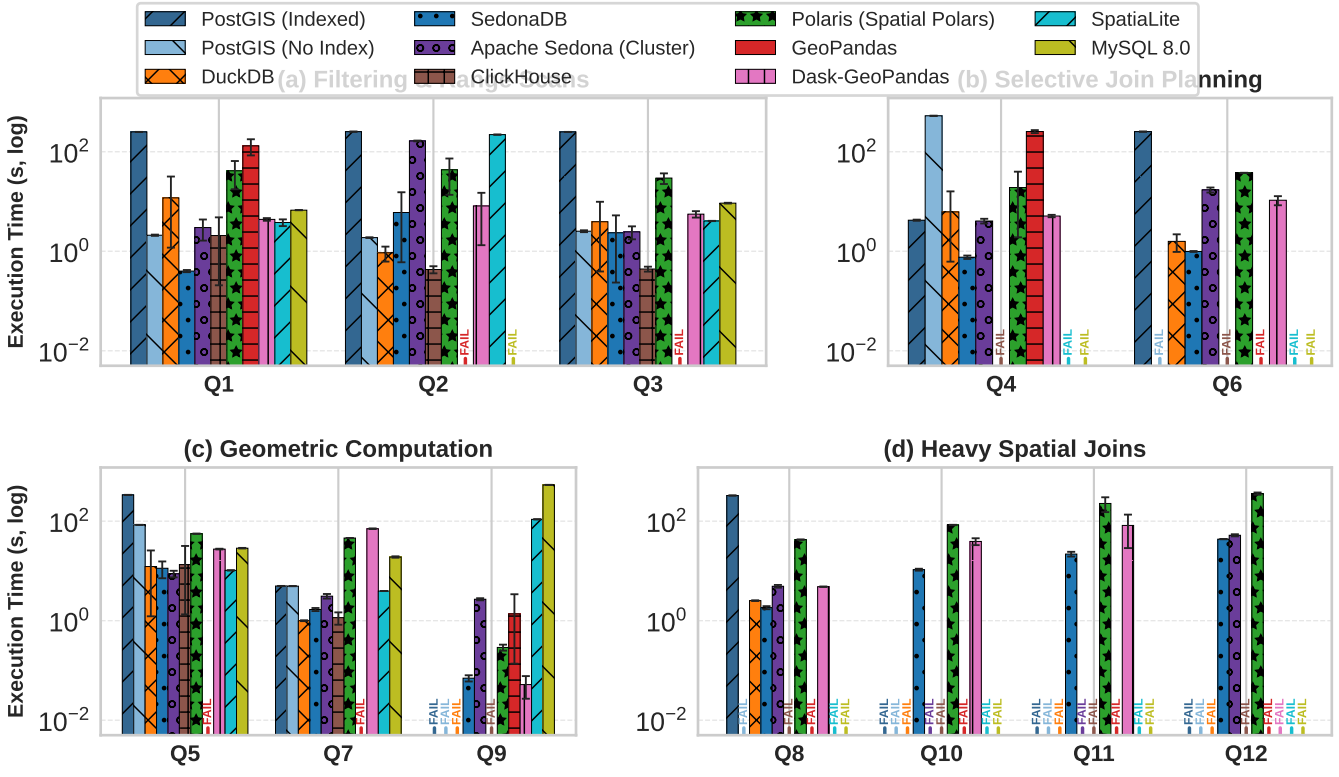

PostGIS (Indexed)
PostGIS (No Index)
DuckDB
SedonaDB
Apache Sedona (Cluster)
ClickHouse
Polaris (Spatial Polars)
GeoPandas
Dask-GeoPandas
SpatiaLite
MySQL 8.0
(a) Filtering Large Scans
(b) Selective Join Planning
(c) Geometric Computation
(d) Heavy Spatial Joins
Execution Time (s, log)
$10^{2}$
$10^{0}$
$10^{-2}$
FAIL
Q1
Q2
Q3
Q4
Q6
Q5
Q7
Q9
Q8
Q10
Q11
Q12